

CSA1708-ARTIFICIAL INTELLIGENCE LAB EXPERIMENTS

Program 5 : Python Program for Missionaries and Cannibal Problem

Aim

To develop a Python program that moves missionaries and cannibals across the river safely without the cannibals outnumbering the missionaries.

Algorithm

- Read the number of missionaries and cannibals on the starting bank.
- Display the initial state.
- Move 1: take 2 cannibals across the river and update the state.
- Move 2: take 2 missionaries across the river and update the state.
- Move 3: bring 1 missionary and 1 cannibal back and update the state.
- Display the state after every move.
- Print "Goal Reached" when all the moves are completed.

Program

```
m = int(input("Enter missionaries: "))
c = int(input("Enter cannibals: "))

print("Initial State:", m, c)

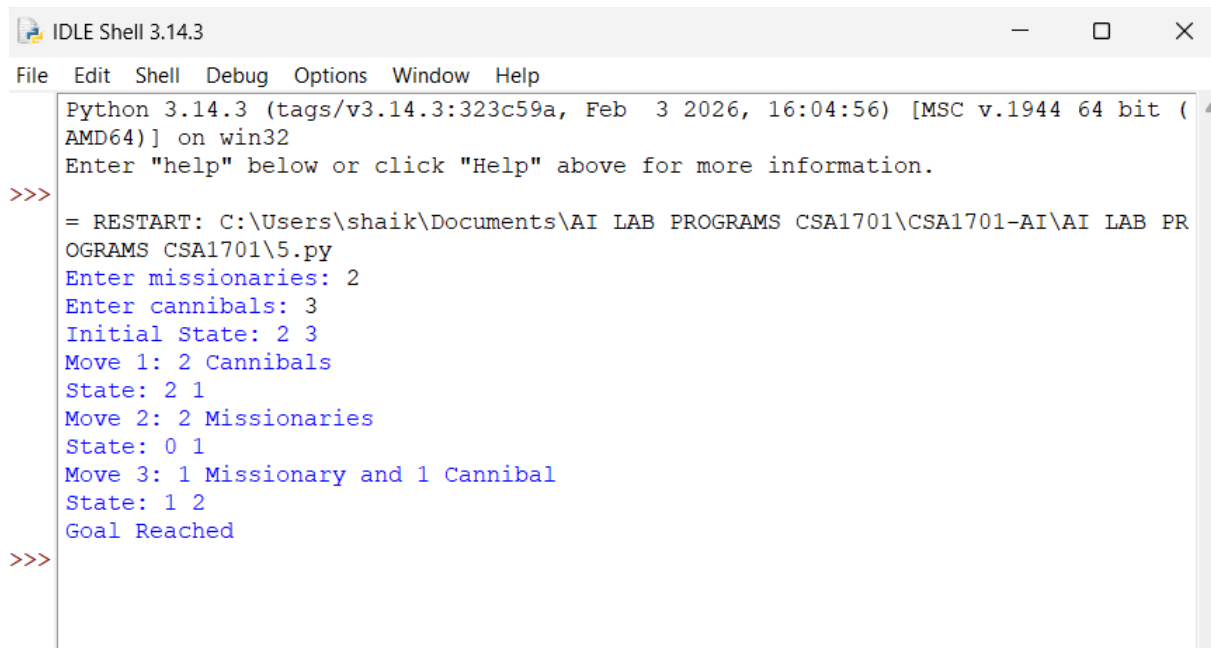
print("Move 1: 2
Cannibals") c -= 2
print("State:", m, c)

print("Move 2: 2
Missionaries") m -= 2
print("State:", m, c)

print("Move 3: 1 Missionary and 1
Cannibal") m += 1 c += 1 print("State:", m,
c)

print("Goal Reached")
```

Output



```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> = RESTART: C:\Users\shaik\Documents\AI LAB PROGRAMS CSA1701\CSA1701-AI\AI LAB PR
OGRAMS CSA1701\5.py
Enter missionaries: 2
Enter cannibals: 3
Initial State: 2 3
Move 1: 2 Cannibals
State: 2 1
Move 2: 2 Missionaries
State: 0 1
Move 3: 1 Missionary and 1 Cannibal
State: 1 2
Goal Reached
>>>
```

RESULT

Thus, the Missionaries and Cannibal Problem was solved successfully using Python.